1. General Features of Inflammation

Definition

Inflammation = tissue response to cell injury and infection, involving delivery and activation of inflammatory cells and host defence molecules from the circulation to remove cause (microbes, toxins) and consequences (necrotic cells) of injury.

5 “R”s of Inflammatory Reaction

1. Recognition of injurious agent
   - o Receptors on cells detect microbes/damage □ production of inflammatory mediators.
2. Recruitment of leukocytes and plasma proteins to the site.
3. Removal of offending agent by phagocytes.
4. Regulation/termination of response once objective achieved.
5. Repair of damaged tissue by regeneration and/or scarring.

Physiologic role: Protective, survival response (vessels + leukocytes + antibodies + complement).

Pathologic aspect: Local tissue damage, pain, loss of function; misdirected/inadequate control □ acute/chronic inflammatory diseases (autoimmune, allergy, metabolic, degenerative, neoplastic).

Acute vs Chronic Inflammation

Feature	Acute	Chronic
Time course	Rapid onset (min–hrs), short duration	Days to years; often follows acute or de novo
Stimulus	Readily eliminated agents	Persistent agents; immune dysregulation, toxins
Cells	Neutrophils + oedema (exudate)	Macrophages, lymphocytes, plasma cells
Tissue injury	Usually limited	More destruction, fibrosis
Clinical signs	4 cardinal signs: rubor, calor, tumor, dolor + loss of function	Local/systemic signs often less obvious; loss of function may be present

Causes of Inflammation

- Infections (bacteria, viruses, fungi, parasites, toxins).
- Tissue necrosis (ischaemia, trauma, chemical/thermal injury).
- Foreign bodies (exogenous or endogenous e.g. urate crystals).
- Immune reactions/hypersensitivity (autoimmune disease, allergy).

Recognition of Microbes & Damaged Cells

- TLRs and other microbial receptors: membrane, endosomal, cytosolic; on epithelial cells, dendritic cells, macrophages, leukocytes □ cytokines and mediators.
- Damage sensors (e.g. NLRs): detect uric acid, ATP, □K^+ □ activate

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Inflammation
Derived from the Latin word inflammatio (nominative) or inflammationem (accusative), meaning "a kindling" or "a setting on fire." This comes from the verb inflammare, meaning "to set on fire" or, figuratively, "to rouse or excite," which clearly relates to the heat and redness (flamma is Latin for 'flame') observed in the process.

inflammasome □ IL 1 production □ leukocyte recruitment.
• Fc and complement receptors on leukocytes: bind opsonised microbes □ phagocytosis, inflammation.
• Circulating proteins: complement, mannose binding lectin, collectins
2. Acute Inflammation
2.1 Vascular Changes
Exudate vs Transudate
• Exudate: Protein rich, cell rich fluid (implies □vascular permeability).
o Pus: purulent exudate rich in neutrophils + necrotic debris ± microbes.
• Transudate: Low protein, few cells; due to hydrostatic/osmotic imbalance without □ permeability.
• Oedema: Excess interstitial or serous cavity fluid (can be exudate or transudate).
Vasodilatation and Stasis
• Early change: arteriolar dilation, opening of capillary beds (mediated esp. by histamine).
□ □blood flow □ heat and redness (erythema).
• Fluid loss □ □RBC concentration □ stasis (slow blood flow, vascular congestion).
□ leukocyte margination along endothelium.
Increased Vascular Permeability
• Endothelial contraction (“immediate transient response”):
o Mediators: histamine, bradykinin, leukotrienes.
o Rapid, short lived (15–30 min) or delayed (e.g. sunburn).
o Opens interendothelial gaps (mainly postcapillary venules).
• Endothelial injury:
o Direct damage (burns, microbes, toxins, neutrophil mediated) □ necrosis, detachment.
o More severe, longer lasting leakage.
• Result: protein/leukocyte rich exudate to tissues.

Exudate
Comes from the verb "to exude," which is derived from the Latin exs□d□re. Exs□d□re is a compound of ex- ("out of") and s□d□re ("to sweat"), literally meaning "to sweat out" or "to ooze out."

Transudate
The noun form, transudation, is a borrowing from the Latin tran(s)s□d□tio. It is formed by combining the prefix trans- ("across," or "through") with s□d□re ("to sweat"), suggesting a process of passing fluid through a membrane.

stasis
This term comes directly from the Greek word (stasis), which literally means "a standing still. In a medical context, it refers to the cessation or slowing of the flow of blood or other body fluids.

Lymphatic Response

• □Lymph flow carrying fluid, leukocytes, debris, microbes.

• May cause lymphangitis (inflamed lymphatics) and lymphadenitis (enlarged, painful draining nodes).

2.2 Leukocyte Recruitment

Key steps

1. Margination, rolling, adhesion
   o Stasis □ leukocytes move to periphery and roll along endothelium.
   o Selectins (L , E , P selectin) mediate low affinity rolling via sialylated glycoprotein ligands.
   o Integrins on leukocytes change to high affinity state under chemokines (e.g. TNF, IL 1) and bind endothelial ligands □ firm adhesion.
2. Transmigration (diapedesis)
   o Mainly in postcapillary venules via interendothelial gaps.
   o Mediated by molecules at junctions (e.g. CD31/PECAM 1).
   o Leukocytes then cross basement membrane (collagenases) to interstitium.
3. Chemotaxis
   o Movement along chemotactic gradient.
   o Exogenous: bacterial products.
   o Endogenous: chemokines (e.g. IL 8), complement C5a, leukotriene B4, others.
   o Binding to receptors □ cytoskeletal changes and directional movement.

Cell type sequence

• 6–24 h: Neutrophils predominate (more numerous, rapid response, firm adhesion; short lived).

• 24–48 h onwards: Monocytes/macrophages dominate (longer survival, proliferate in tissues).

• Exceptions: lymphocytes in viral infections; eosinophils in parasites/allergy, etc.

2.3 Phagocytosis and Leukocyte Mediated Injury

Phagocytosis Steps

1. Recognition and

Chemotaxis
A compound term formed from the combining forms chemo- and -taxis. Chemo- relates to "chemical action" or "chemicals," while -taxis comes from the Greek (taxis), meaning "arrangement" or "order." It thus means "arrangement/movement in relation to chemicals."

Phagocytosis
This is a compound word formed from Ancient Greek roots. It combines (phagein), meaning "to eat," and (kytos), meaning "cell," giving the literal meaning "cell-eating."

attachment
o Receptors: mannose, scavenger, integrins.
o Opsonins (IgG, C3b, certain lectins) greatly enhance binding via specific receptors.
2. Engulfment
o Pseudopods surround particle → phagosome.
o Fusion with lysosome → phagolysosome.
o Some lysosomal contents may leak extracellularly → tissue damage.
3. Intracellular killing
o ROS (NADPH oxidase "respiratory burst" + MPO → HOCl) damage lipids, proteins, nucleic acids.
o NO/Reactive nitrogen species (iNOS in activated macrophages) combine with superoxide → toxic radicals.
o Lysosomal enzymes: hydrolases, proteases, defensins, lysozyme, etc. degrade microbes and debris.
Leukocyte Mediated Tissue Injury
• Same mechanisms that kill microbes can injure host tissues:
o Difficult to eradicate infections (e.g. TB).
o Autoimmune diseases.
o Allergic reactions.
• "Frustrated phagocytosis": large surfaces (e.g. immune complexes on GBM, urate crystals) → extracellular release of granule contents.
• Control mechanisms: antiproteases (e.g. α1 antitrypsin), antioxidants, etc.
Termination of Acute Inflammation
• Removal of stimulus; mediators no longer produced; many are short lived.
• Neutrophil apoptosis in tissues.
• Active "switch off":
o Change from leukotrienes (pro inflammatory) to lipoxins (anti inflammatory).
o Release of anti inflammatory cytokines (e.g. TGF β, IL 10).
2.4 Chemical Mediators of Acute Inflammation
General principles
• Produced only after stimulation; short lived.
• May be cell derived (stored or newly synthesised) or plasma derived (from liver, circulating as inactive precursors).
• One mediator can trigger others (cascades, amplification).
Key Mediators (high yield)
• Vasoactive amines
o Histamine: mast cells, basophils, platelets; released by injury, IgE crosslinking, C3a/C5a.
→ Arteriolar vasodilatation + ↑venular permeability.
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Arachidonic Acid (AA) Metabolites (Eicosanoids)
o Derived from membrane phospholipids (phospholipase A2).
o COX pathway □ prostaglandins
□ PGD_2, PGE_2: vasodilation, □permeability; PGE_2 also pain, fever.
□ PGI_2: vasodilator, inhibits platelet aggregation.
□ TxA_2: vasoconstrictor, promotes aggregation.
o Lipoxygenase pathway □ leukotrienes
□ LTC_4, LTD_4, LTE_4: □permeability, bronchospasm.
□ LTB_4: strong neutrophil chemoattractant, activation.
o Lipoxins: anti inflammatory (inhibit neutrophil recruitment).
• Cytokines and Chemokines
o TNF, IL 1 (mainly from macrophages):
□ Local: endothelial activation (adhesion molecules, cytokines, procoagulant), leukocyte/fibroblast activation.
□ Systemic: fever, acute phase response, shock.
o Chemokines: chemotaxis and activation of specific leukocytes (e.g. IL 8 for neutrophils).
• Complement system
o Activation via classical, alternative, or lectin pathways □ C3 convertase □ C3a/C3b; C5 convertase □ C5a/C5b □ MAC (C5b–C9).
o Functions:
□ C3a, C5a: anaphylatoxins (histamine release); C5a also chemotactic/activating.
□ C3b: opsonin.
□ MAC: cell lysis (especially Neisseria).
o Host protection: C1 inhibitor, DAF, CD59, Factor H.
• Others
o Platelet activating factor: platelet aggregation, vasodilation, □permeability, bronchoconstriction.
o Kinins (bradykinin): pain, vasodilation, □permeability, smooth muscle contraction.
Pharmacologic Inhibitors (important clinically)
• Antihistamines (H1 antagonists).
• NSAIDs (non selective COX inhibitors): □prostaglandins (analgesic, antipyretic, anti inflammatory).
• Selective COX 2 inhibitors: fewer gastric side effects but □thrombosis risk.
• Corticosteroids: broad suppression (□COX 2, PLA2, cytokines, iNOS).
• Leukotriene inhibitors/receptor antagonists: useful in asthma.
• Biologic antagonists of TNF, IL 6, IL 17 for chronic inflammatory

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arachidonic
Information currently unavailable

diseases.

2.5 Morphologic Patterns & Outcomes of Acute Inflammation

Morphologic Patterns

- Serous inflammation: thin, cell poor fluid in cavities or blisters (e.g. skin blister, pleural effusion).
- Fibrinous inflammation: fibrin rich exudate on serosal surfaces (pericardium, pleura, meninges).
  - If not removed → organisation → fibrosis, adhesions.
- Suppurative/purulent inflammation: pus (neutrophils, necrotic debris, oedema); typical of pyogenic bacteria.
  - Abscess: localised pus collection with central necrosis surrounded by neutrophils, granulation tissue, fibrosis.
- Ulcer: local loss of epithelium/epidermis and underlying tissue (e.g. peptic ulcer; skin ulcers).

Outcomes

1. Complete resolution (ideal) – when injury is mild, short lived, tissue can regenerate, exudate cleared.
2. Healing by scarring/fibrosis – substantial tissue destruction, non regenerative tissues, persistent fibrin.
3. Progression to chronic inflammation – persistent stimulus or interference with normal healing.

3. Chronic and Granulomatous Inflammation

Definition

Prolonged inflammation (weeks–months) with coexistence of active inflammation, tissue destruction, and attempts at repair.

Causes

- Persistent infections (e.g. TB, certain fungi, parasites) → often granulomatous.
- Immune mediated diseases (autoimmune, chronic infections, IBD).
- Prolonged exposure to toxic agents (silica, lipids in atherosclerosis).

Morphology

- Mononuclear cell infiltrate (macrophages, lymphocytes, plasma cells).
- Tissue destruction.
- Healing attempts: angiogenesis, fibrosis.

Key Cells and Pathways

Macrophages

- Central effector: phagocytosis, mediator and growth factor secretion, antigen presentation.

Granulomatous
This adjective is based on the noun "granuloma," which comes from the Latin grānulum, meaning "small grain" (a diminutive of grānum or 'grain'). The suffix -oma is a Greek-derived medical suffix used to denote a mass, tumor, or morbid growth, resulting in a term for a mass made of small granules.

ROS, NO, IL 1, IL 12, IL 23).
o Alternative (M2): IL 4, IL 13 → tissue repair, fibrosis, anti inflammatory (TGF β, IL 10, growth factors).
Lymphocytes
• $CD4^+$ T cells:
o Th1 → IFN γ → M1 activation.
o Th2 → IL 4, IL 5, IL 13 → eosinophils + M2.
o Th17 → IL 17 → chemokines → neutrophil/monocyte recruitment.
• T cells and macrophages amplify each other (positive feedback).
Other Cells
• Eosinophils (IgE mediated, parasitic; major basic protein toxic to parasites and host epithelium).
• Mast cells (immediate hypersensitivity; present in both acute and chronic).
• Neutrophils may persist in “acute on chronic” patterns.
Granulomatous Inflammation
Definition: Focal collection of activated macrophages (epithelioid cells) ± giant cells, rimmed by lymphocytes ± fibrosis.
Purpose: Attempt to contain difficult to eradicate agents.
Types
• Foreign body granulomas: around inert, non immunogenic material (talc, sutures), no strong T cell response.
• Immune granulomas: persistent T cell mediated response to microbes or self antigens (e.g. TB, syphilis, Crohn disease).
• Sarcoid granulomas: non caseating, “naked” (few lymphocytes), unknown cause.
Morphology
• Epithelioid histiocytes, multinucleated giant cells.
• Central caseous necrosis typical of TB (cheese like).
• Non caseating in Crohn disease, sarcoidosis, foreign body.
• Outcome: fibrosis; may be extensive.
4. Systemic Effects of Inflammation
Acute Phase Response
Mediated mainly by cytokines (TNF, IL 1, IL 6, interferons).
• Fever:
o Exogenous pyrogens (LPS) → endogenous pyrogens (IL 1, TNF) → ↑COX → PGE_2 in hypothalamus → reset set point → vasoconstriction, shivering, ↑heat production.
• Acute phase proteins (from liver): CRP, fibrinogen, SAA.
o Opsonisation, complement activation; fibrinogen ↑ESR.
o Chronic elevation can cause complications (e.g. secondary amyloidosis; CRP as risk marker in CAD).
• Leukocytosis:
o Early: accelerated release (± left shift).
o Later: marrow proliferation (CSFs).
o Patterns: neutrophilia (bacterial), lymphocytosis (viral), eosinophilia

• Other signs: tachycardia, hypertension, chills, rigors, anorexia, malaise, somnolence, □sweating.

Septic Shock (SIRS in sepsis)

• Severe systemic response to bacterial infection: high LPS/bacterial products → massive TNF/IL 1.

• Features: DIC, hypotensive shock, metabolic disturbances (insulin resistance, hyperglycaemia); high mortality.

5. Tissue Repair: Regeneration and Scarring

Definition

Restoration of tissue architecture and function after injury via regeneration and/or scarring.

5.1 Regeneration

Mechanism

• Proliferation of residual differentiated cells and/or activation of tissue stem cells.

• Requires: preserved ECM scaffold + growth factors + integrin–ECM signalling.

Tissue Proliferative Capacity

• Labile tissues: continuously dividing (epidermis, GI mucosa, haematopoietic cells) → excellent regeneration if stem cells intact.

• Stable tissues: quiescent but can re enter cycle when needed (parenchyma of liver, kidney; fibroblasts, endothelial, smooth muscle) → limited regeneration (liver is an exception with high capacity).

• Permanent tissues: terminally differentiated (neurons, cardiomyocytes) → damage usually repaired by scarring.

Example: Liver Regeneration

• Up to ~90% of liver can regenerate after partial hepatectomy:

o Priming (IL 6) → hepatocytes sensitive to growth factors.

o Growth phase (HGF, TGF α) → hepatocyte proliferation.

o Termination (TGF β family) → return to quiescence.

• If hepatocyte proliferation impaired → progenitor cells (canals of Hering) may contribute.

5.2 Scarring (Fibrosis)

When occurs

• Tissue incapable of full regeneration or ECM framework destroyed.

• Aim: structural stability rather than complete restoration.

Sequential Steps

1. Haemostasis and inflammation – platelet plug, fibrin, acute inflammation.

Regeneration
Comes from the Latin regener□ti□, which is built from the prefix re- ("again" or "anew") and the root gener□ti□ ("generation" or "production"). The entire term literally means "to bring forth again" or "generate anew."

2. Granulation tissue formation
o Proliferating fibroblasts + new thin walled vessels + residual inflammation.
3. Connective tissue deposition
o Fibroblast migration/proliferation; collagen and ECM synthesis; driven mainly by M2 macrophages (TGF β, growth factors).
4. Remodelling
o Balance of ECM synthesis vs degradation by MMPs (activated at injury sites) and inhibited by TIMPs → final scar.
Angiogenesis (within granulation tissue)
• Vasodilation and ↑permeability (NO, VEGF).
• Pericyte separation and BM degradation (Ang1/2, proteases).
• Endothelial migration and proliferation (VEGF, FGF).
• Maturation: recruitment of pericytes/smooth muscle (PDGF), BM deposition and stabilisation (TGF β), vessel regression.
5.3 Factors Affecting Repair
Local
• Infection (most important cause of delayed healing).
• Mechanical stress (movement, tension, pressure).
• Poor perfusion (PVD, diabetes, venous congestion).
• Foreign bodies.
• Nature, site, and extent of injury (permanent tissue → inevitable scar; cavity exudates → resolution vs organisation).
Systemic
• Diabetes mellitus.
• Nutritional status (protein, vitamin C deficiency).
• Glucocorticoids (impair TGF β, may weaken scar but sometimes therapeutically useful).
• Age (younger → better healing).
5.4 Abnormalities in Tissue Repair
• Deficient scar formation: chronic wounds, ulcers, wound dehiscence (obesity, infection, poor nutrition).
• Excessive scarring:
o Hypertrophic scar (raised, within wound, regresses).
o Keloid (beyond wound margins, does not regress).
o Excess granulation tissue ("proud flesh") → blocks re epithelialisation.
• Contractures: exaggerated wound contraction → deformities, restricted movement (common after burns).
5.5 Examples
Cutaneous Wound Healing
• First intention (primary union): clean, closely apposed incisions, minimal cell death.
o Rapid epithelial bridging; limited granulation tissue; small scar; tensile strength ↑ over months (max ~70–80% of normal).

o More inflammation, granulation tissue, fibrosis; significant wound contraction via myofibroblasts.

Fibrosis in Parenchymal Organs

• Mechanism analogous to dermal scarring.

• Clinical consequences more serious due to loss of functional parenchyma (e.g. cirrhosis, interstitial lung fibrosis).